(a) Median max force across 639 candidates

Candidate stage

Stable early (n=207) 2.6 5.3 2.6 2.4 3.1 2.9 2.7 2.7 2.7 2.7 2.7 2.7

Stable pre-onset (n=216) 2.6 5.6 2.6 2.5 3.2 3 2.8 2.8 2.8 2.8 2.8 2.8

Precursor (n=208) 3.7 12 3.2 17 30 31 15 18 17 16 15 17

Threshold / terminal (n=7) 1.3e+02 1.7e+02 3.6e+02 4.5e+02 8.9e+02 1.3e+03 8.9e+02 4.9e+02 9.2e+02 1.6e+03 3.3e+03 2.4e+03

Stable terminal (n=1) 2.6 13 2.7 2.4 3 2.7 2.8 2.8 2.7 2.8 2.8 2.8

Lmax2 / seed 123
Lmax2 / seed 234
baseline / seed 345
OMAT legacy
MPA-0 medium
MP-0b3 medium
FT-600K / 123
FT-600K / 234
FT-600K / 345
FT-MultiT / 123
FT-MultiT / 234
FT-MultiT / 345

Median max force (eV/A)
10^3
10^2
10^1

(b) Model force distributions by candidate stage

Max force (eV/A)
10^4
10^3
10^2
10^1

Stable early (n=207)
Stable pre-onset (n=216)
Precursor (n=208)
Threshold / terminal (n=7)
Stable terminal (n=1)

Candidate stage

From Scratch
Foundation / zero-shot
FT-600K
FT-MultiT

arm 1
arm 2
arm 3
100 eV/A

(c) Candidate and reference embedding

Model arm

PCA 2
200
150
100
50
0

Near-origin detail
0.0
−2.5
−5.0
−7.5
−10.0
−12.5
−15.0
−1
0
1
2
3

−20
−15
−10
−5
0

PCA 1

Train
Valid
Test
NEB reference
Stable early (n=207)
Stable pre-onset (n=216)
Precursor (n=208)
Threshold / terminal (n=7)
Stable terminal (n=1)

(d) Distribution shift versus model disagreement

Across-model max-force range (eV/A)
10^4
10^3
10^2
10^1
10^0

10^1
1.1×10^1
1.2×10^1
1.3×10^1
1.4×10^1

Distance to training set (feature units)

Stable early (n=207)
Stable pre-onset (n=216)
Precursor (n=208)
Threshold / terminal (n=7)
Stable terminal (n=1)